

very (VERY) large scale knowledge representation

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Contributions from many people
in the KR&R group over many years.

And thanks to NWO for
a 750k€ TOP grant for this

Very (VERY) large scale KR:
Knowledge Graphs

What's it like to be a computer ?

林克昌 根留台灣 可能增高

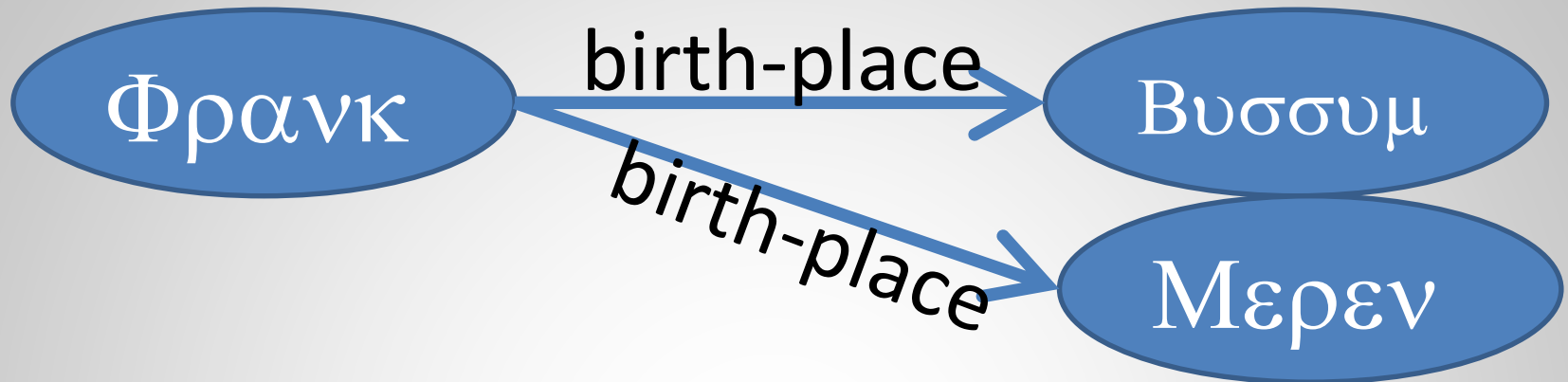
在愛戴者熱心奔走之下，華裔名指揮家林克昌根留台灣的可行性又提升了幾分。兩廳院主任李炎、國家音樂廳樂團副團長黃奕明日前親赴林克昌、石聖芳寓所拜會，並提出多場客席邀約。此外，台灣省立交響樂團團長陳澄雄也早早「下訂」，邀請林克昌赴台中霧峰，從八月十日起訓練省交，為期長達一個月。

在台灣諸多公家樂團中，陳澄雄是以實際行動表達對林克昌肯定的樂界人士之一，曾多次公開表示對林克昌指揮才華的欽佩，而且幾乎每個樂季都邀請林克昌客席演出。

此外，林克昌上個月赴俄羅斯與頂尖的「俄羅斯國家管絃樂團」灌錄了柴可夫斯基晚期三大交響曲以及「羅密歐與茱麗葉」、「斯拉夫進行曲」、「義大利隨想曲」，最後的DAT母帶也在前兩天寄回台灣。製作人楊忠衡與林克昌試聽之後，都對錄音效果—尤其音質表現感到相當滿意，楊忠衡估計呈現了七分林克昌指揮神韻。

俄羅斯國家管絃樂團首席布魯尼日前也讚譽林克昌的指揮藝術有三大特點：一是控制自如的彈性速度；二是強烈的動態對比；三是宛如呼吸歌唱的旋律處理。這些對錄音師而言都構成很大挑戰。俄國錄音師雖然採用多軌混音，但定位、場面都有可觀之處。。

explicit & formal semantics



- Φρανκ is person
- birth-place relates person to location

lowerbound

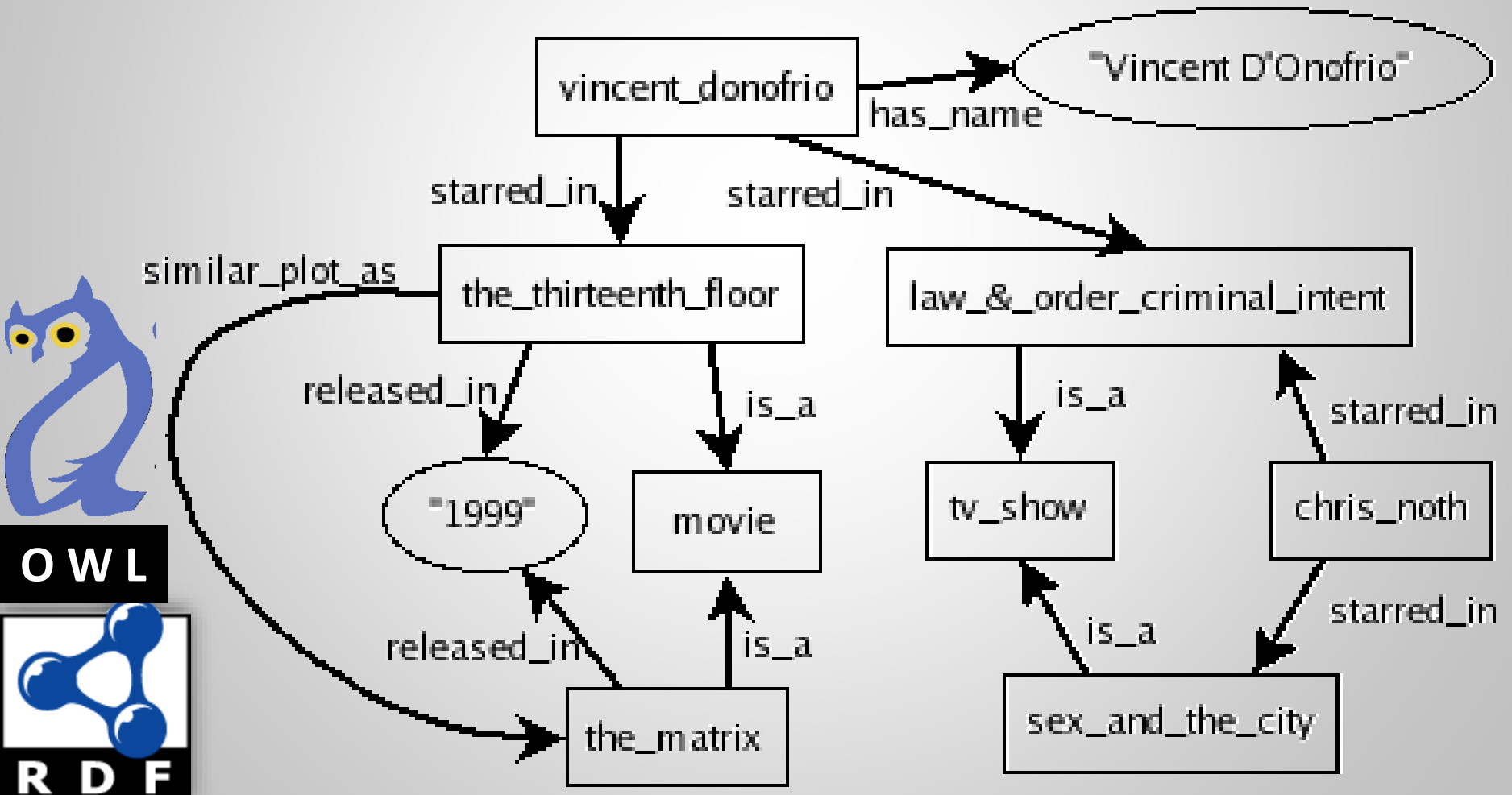
- birth-place relates 1 person to 1 location
- Βυσσυμ = Μερεν

upperbound

Semantics = predictable inference

What are Knowledge Graphs

Most widely used KR formalism in history
(by many orders of magnitude)



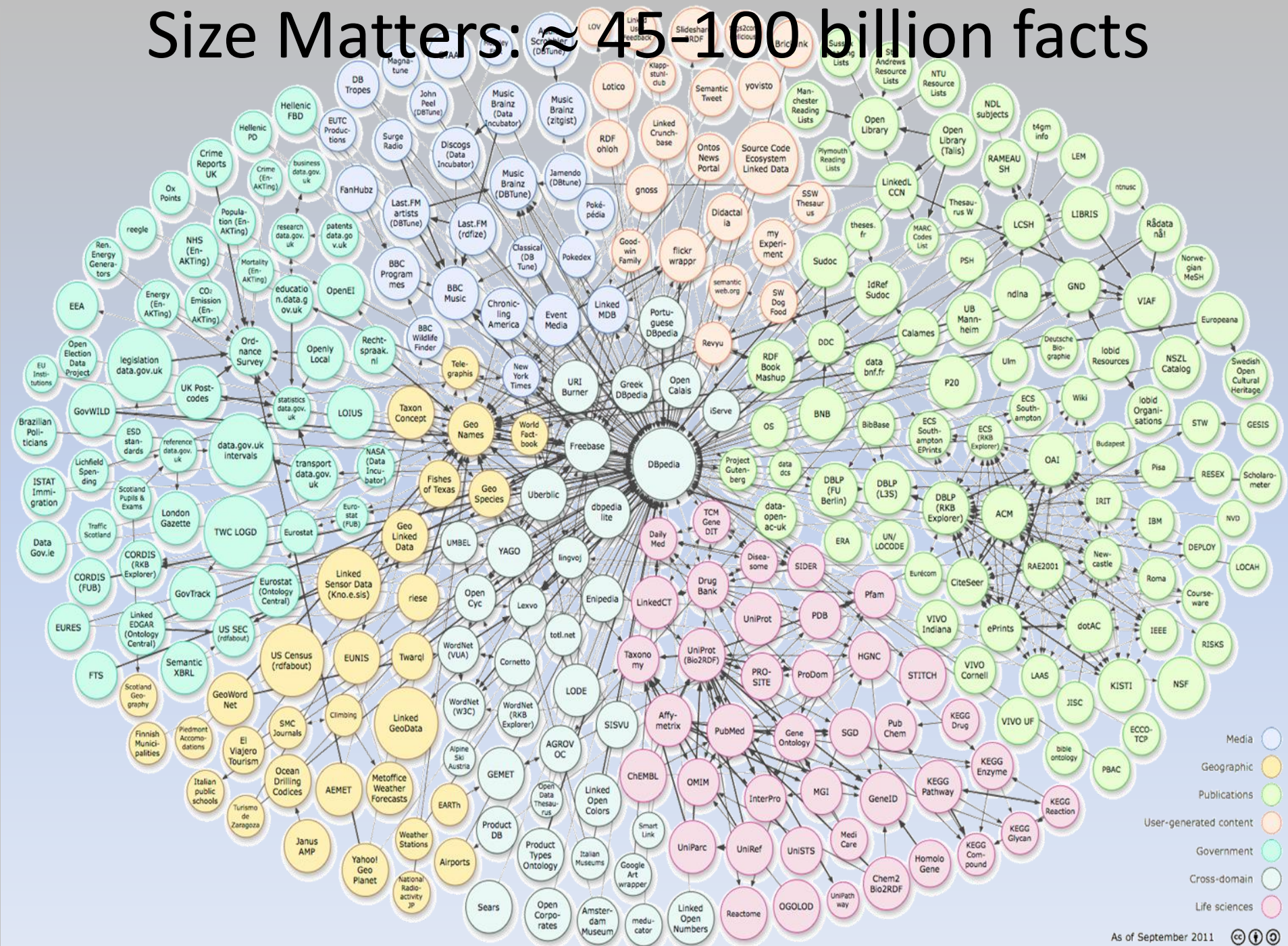
What are Knowledge Graphs (for logicians)

- ground binary predicate: $P(O_1, O_2)$
- Limited existential variables: $\exists x: P(C_1, x) \wedge P(C_2, x)$
- Types is unary predicate: $T_i(x)$
- Subtypes $\forall x: T_1(x) \rightarrow T_2(x)$
- Type restrictions $\forall x, y: P(x, y) \rightarrow T_1(x) \wedge T_2(y)$
- Equality: $O_1 = O_2$
- Extensions to **DL**:
 - Distjointness of types
 - Cardinality restrictions (0,1)
 - always decidable: sub-FOL.

Logical deduction

What's in these knowledge graphs,
and how big do they get?

Size Matters: $\approx 45\text{--}100$ billion facts



How big are Knowledge Graphs?

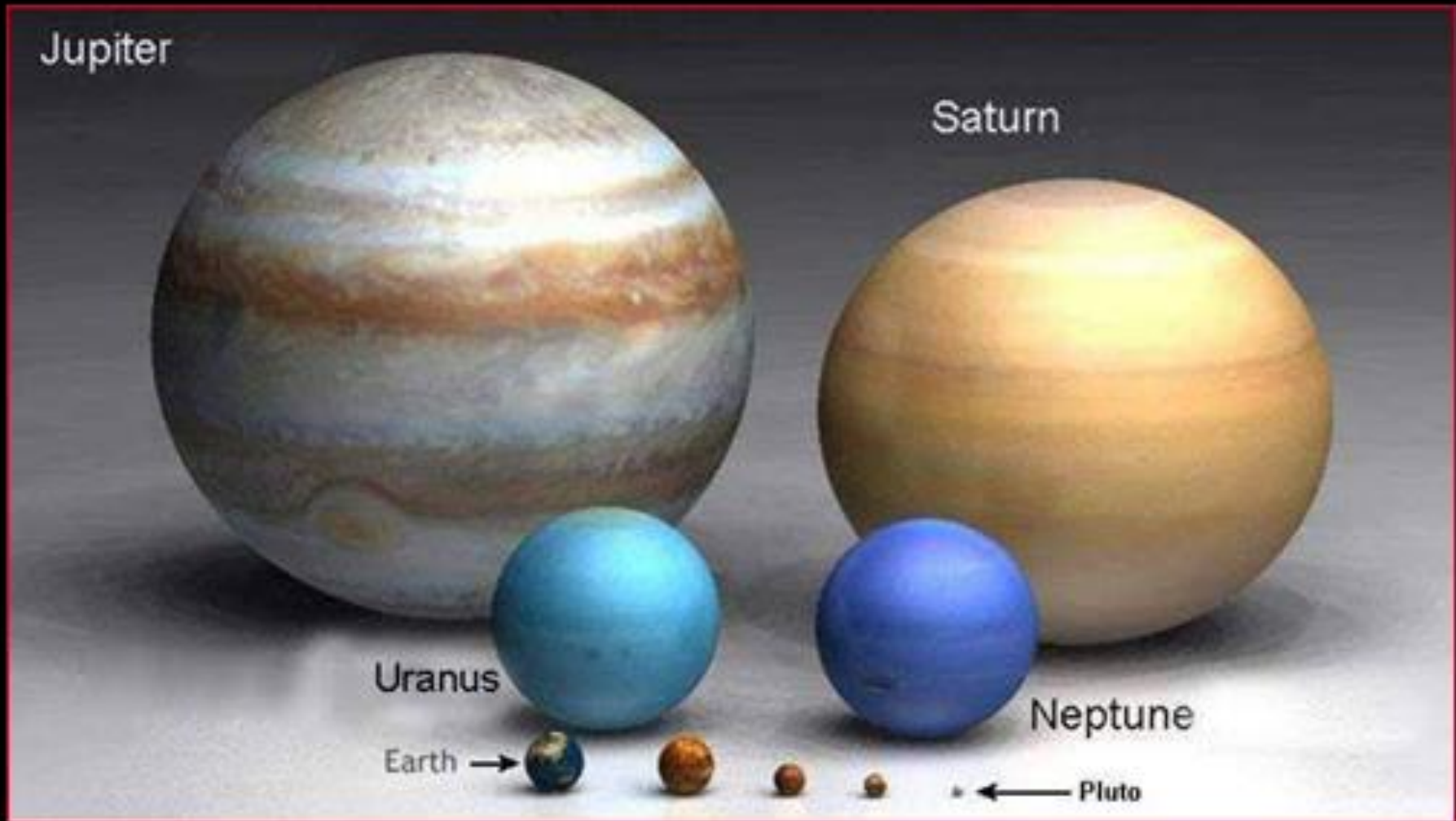
A white golf ball is positioned on a green tee, set against a backdrop of a bright sun in a blue sky and a field of green grass. The text "1 triple" is superimposed on the ball.

1 triple

Size of the current Semantic Web

$\sim 10^{10}$ Triples

Jupiter



≈ 1 triple per web-page

Is anybody using this stuff?

Tencent 腾讯

UniProt USGS

Google
Bing

Alibaba.com

Baidu 百度

PubMed

Uber
eats

Cochrane

EMBL-EBI

facebook

DEUTSCHE
NATIONAL
BIBLIOTHEK

airbnb

ANTONI
VAN
LEEUVENHOEK
FOUNDATION

MAASTRO

The
New York
Times

BBC

LE
ITY

européana

NXP

NETFLIX



REUTERS

Overheid.nl

{BnF

National Library
of Sweden

LIBRARY OF
CONGRESS



Bodleian Libraries
UNIVERSITY OF OXFORD

RENAULT

Bloomberg

legislation.gov.uk

EPA
United States
Environmental Protection
Agency

Wolters Kluwer

IOS
Press

kadaster

POLITIE

European
Commission

Walmart

SIEMENS

LEI
INFO

EUROMONEY

zalando

ebay

BEST
BUY



Deloitte.

accenture

SPRINGER NATURE

amazon.com

ELSEVIER

Web Images Maps Shopping More Search tools

About 158,000,000 results (0.26 seconds)

[Utrecht.nl - Homepage](#)

[www.utrecht.nl/](#) - Translate this page

Utrecht de gemeentesite. Welkom op de website van de Gemeente **Utrecht**. Hier vindt u alle informatie over de stad en de diensten van de Gemeente **Utrecht**.

[Loket Burgerzaken - Contact - Afspraak maken Burgerzaken - Milieu en afval](#)

[Utrecht - Wikipedia, the free encyclopedia](#)

[en.wikipedia.org/wiki/Utrecht](#)

Utrecht city and municipality is the capital and most populous city of the Dutch province of **Utrecht**. It is located in the eastern corner of the Randstad conurbation, ...

[Utrecht University - Treaty of Utrecht - Utrecht - DOS](#)

[Utrecht University](#)

[www.uu.nl/EN/Pages/default.aspx](#)

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[FC Utrecht](#)

[fcutrecht.nl/](#) - Translate this page

FC **Utrecht** maakt gebruik van cookies om u een goede website ervaring te bieden en deze te kunnen verbeteren. Ik accepteer cookies X ...

[Bezoek Utrecht](#)

[www.bezoek-utrecht.nl/](#) - Translate this page

Welkom in **Utrecht**! **Utrecht**, het kloppende hart van Nederland. Een stad met een unieke dynamiek. Klein genoeg om wandelend de middeleeuwse binnenstad ...

Is anybody using this for real?

Schema.org:



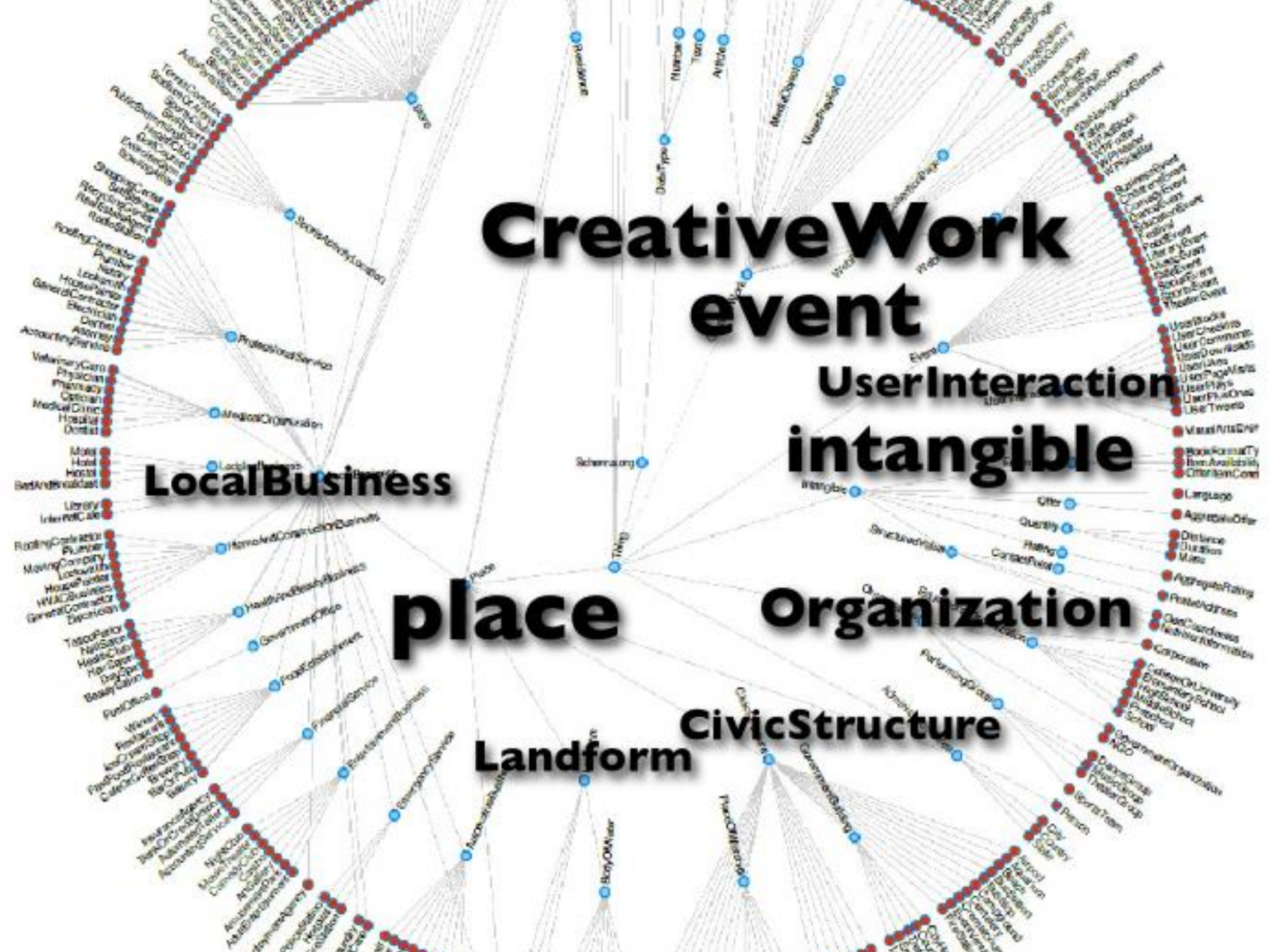
Vocabulary to describe “things on the web”

Agreed upon by all major search engines

600+ types, 1000 properties

used by 100M+ sites

shows up in 36% of all Google results



**CreativeWork
event**

**UserInteraction
intangible**

Organization

CivicStructure

Landform

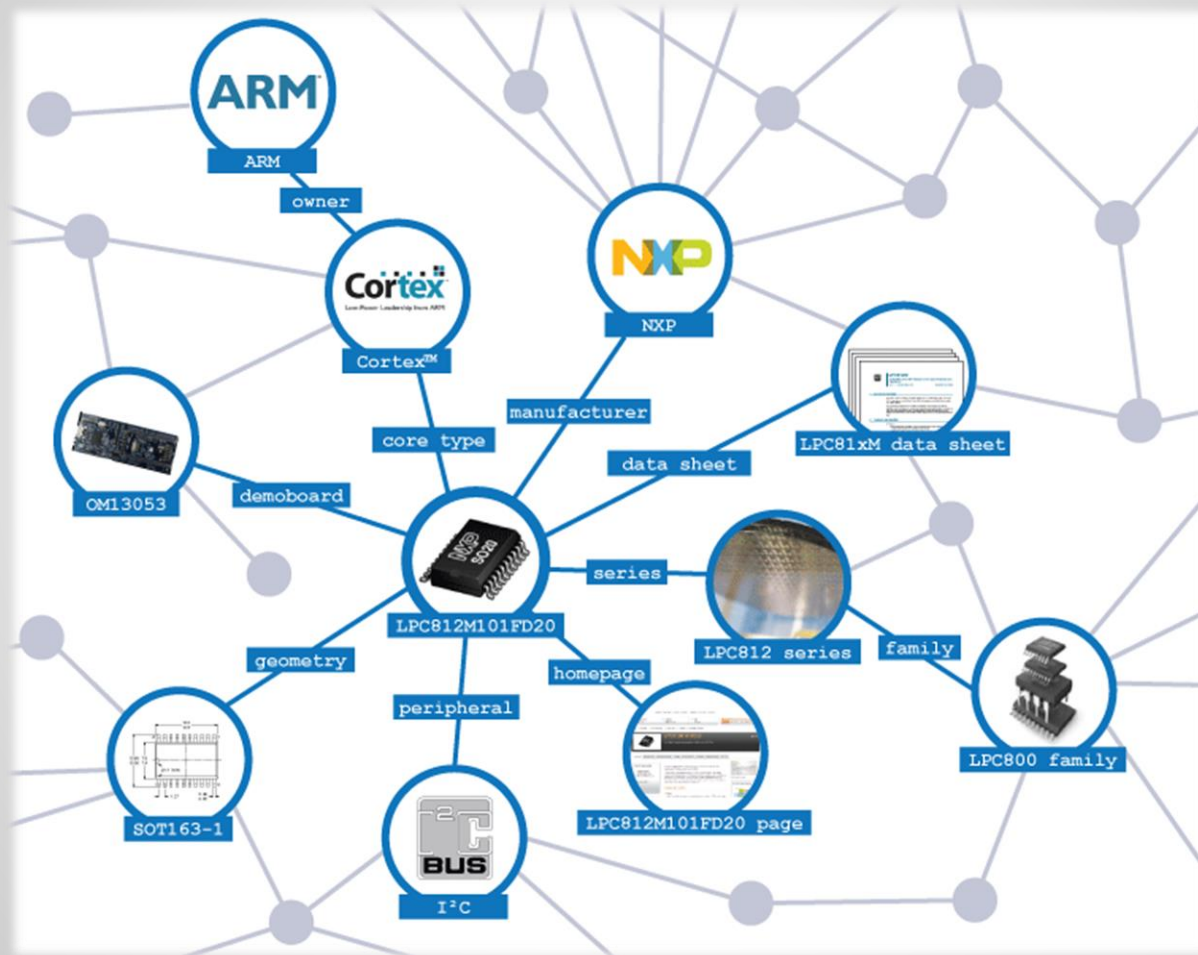
place

LocalBusiness

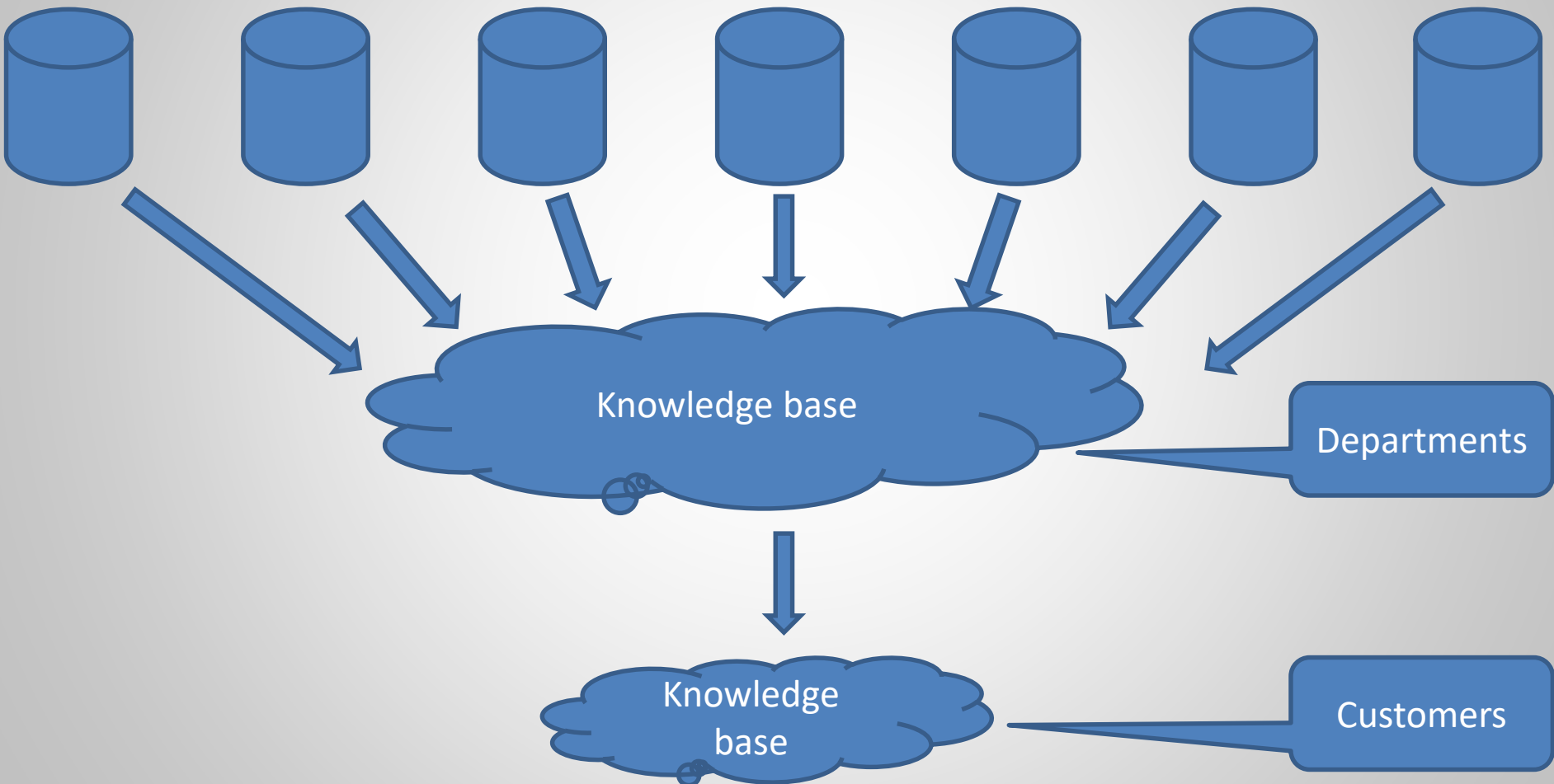
- NXP is a semiconductor (microchip) manufacturer
- Established: 2006 (formerly a division of Philips) with 50+ years of experience in semiconductors
- Headquarters: Eindhoven, The Netherlands
- Customers include Apple, Bosch, Continental, Delphi, Gemalto, Giesecke/Devrient, Huawei, NSN, Panasonic and Samsung
- Portfolio of **26,000+** products



They built their own ontology



NXP: data integration about 26.000 products



SPORT OLYMPICS

The Olympic Broadcaster
London 2012

Home | Football | Formula 1 | Cricket | Rugby Union | Olympics | The Ashes

More sports ▾

Countries | Athlete | Britta Steffen

Schedule & Results | Medals | More Olympic sports ▾

Britta Steffen



Biography

SWIMMING

Britta Steffen (born 16 November 1983 in Schwedt, Germany) is a German swimmer who specializes in freestyle, who is the current holder of the world record in women's 50 and 100 metre freestyle.

[Full biography ▾](#)

Olympic schedule

**414 days to go**

Olympic Games 27 July-12 Aug 2012

[View full olympic schedule](#)

Key facts

Country
GermanyCompetes in
Womans 50m
Freestyle
Womans 100m
FreestyleTeam
Germany Women's
4 x 100m Freestyle
RelayDate of birth
03/03/83Height
1.88m/6'

WOMEN'S 100M FREESTYLE

Britta Steffen sets
new record 100m

Jamaican sprint superstar Usain Bolt claims yet another Olympic record with a stunning victory in the 100m heats.

Olympic venues

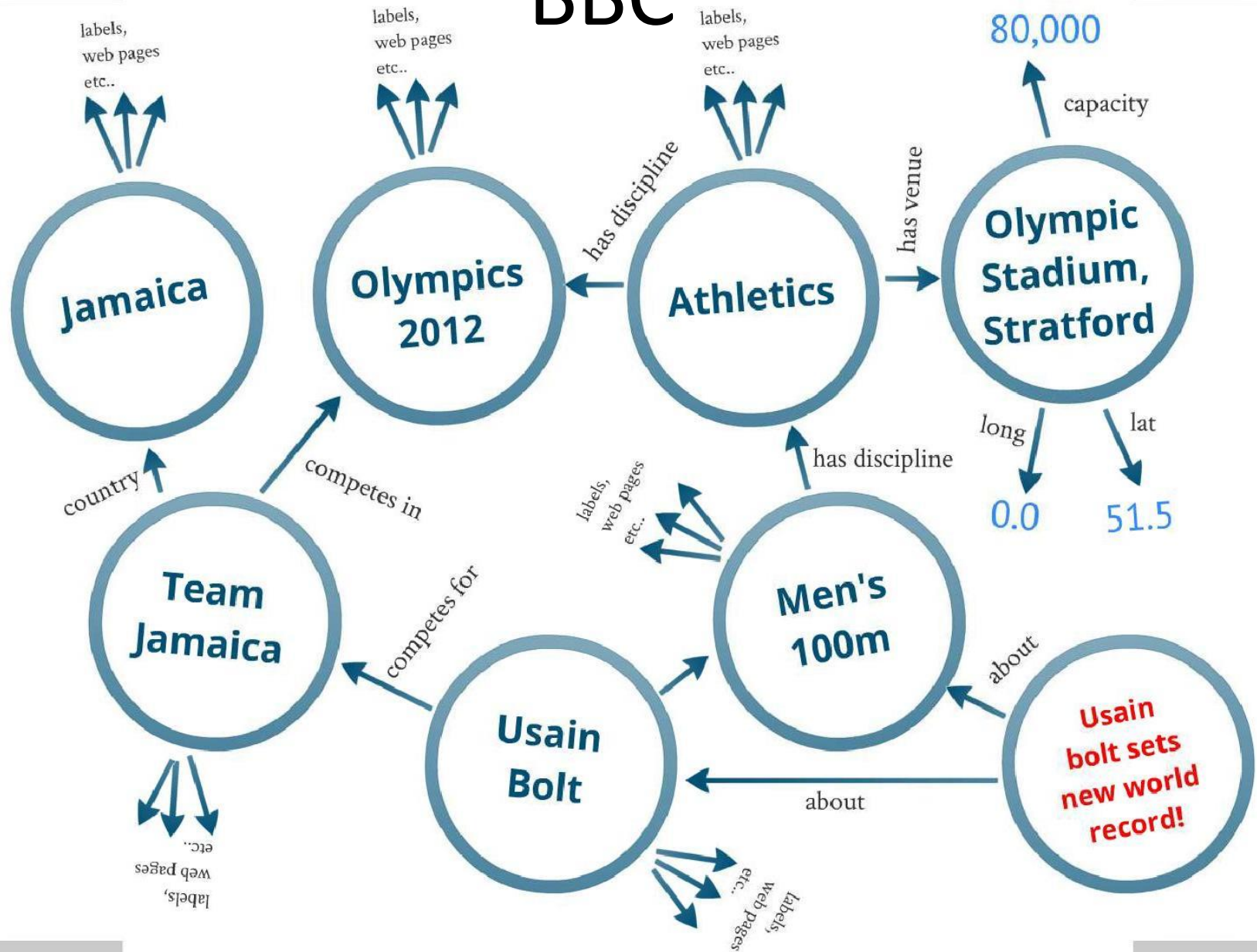


From the 27th of July 2012, 17,000 athletes will compete in 31 venues in the Olympic Park, across London and at out of London venues

[View all London 2012 venues](#)

Olympic 2012 programmes

BBC

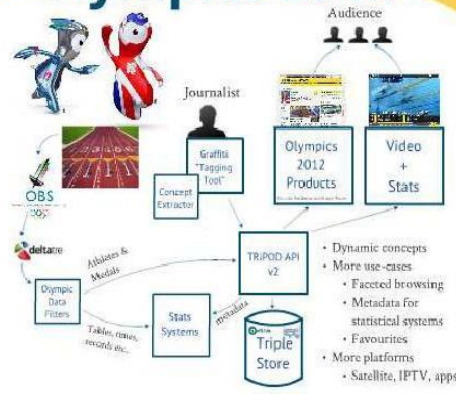


BBC Linked Data Platform

(a brief history...)

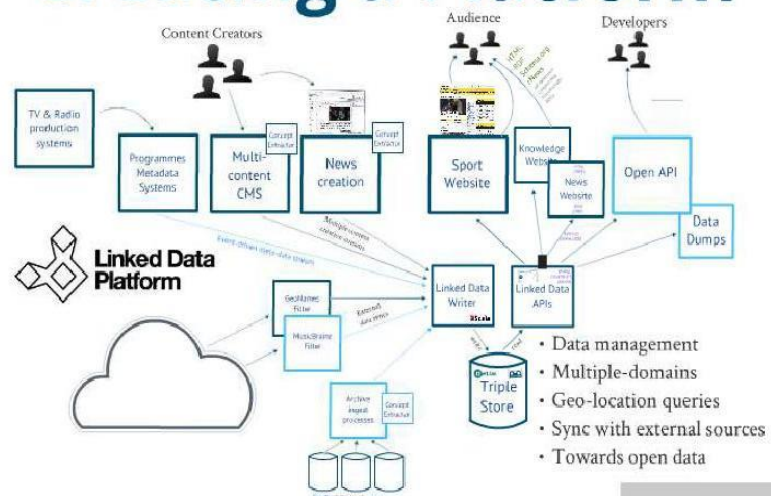
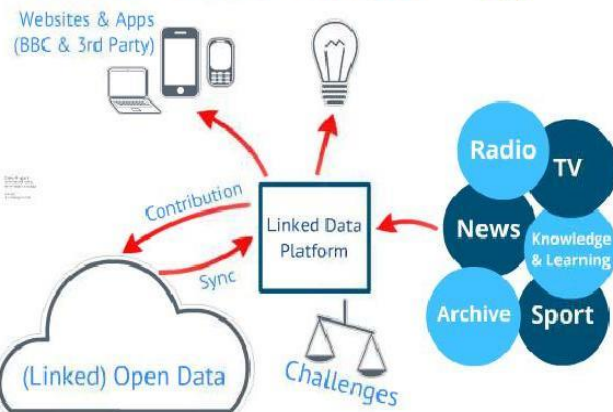
World Cup 2010

Olympics 2012



The future?

Creating a Platform



Government

navigation

- Main Page
- People
- Demos
- Videos
- Tutorials
- Pubs/Presentations
- Labs
- RSS Feeds
- Contact us

linked data catalog

- Data.gov catalog(all)
- Data.gov catalog(ogd)
- Non-Data.gov catalog

maintenance

- Vocabulary
- Recent changes
- Random Page

internal pages

- Internal
- Internal meetings

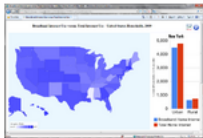
search

toolbox

- What links here
- Related changes
- Special pages
- Printable version
- Permanent link

Demos

Demo: Broadband Internet Use versus Total Internet Use - United States Households, 2009



- **Description:** This demo shows the percentage of households in each state with broadband out of the total number of households with internet access. This is based on the division of broadband into urban and rural areas.
- **Live Demo(s):** [\[1\]](#)
- **Dataset(s):** [Dataset 10040](#)
- **Technology Used:** [RDF](#), [SPARQL](#), [SparqlProxy](#), [Google Visualization API](#)
- **Creator(s):** [Peter Coons](#)

Demo: Clean Air Status and Trends - Ozone



- **Description:** This demo shows a map of the U.S. depicting the Clean Air Status and Trends Network (CASTNET) Ozone monitoring stations and their average Ozone level. Clicking a station shows the time-series of measurements.
- **Live Demo(s):** [\[2\]](#)
- **Dataset(s):** [Dataset 8](#), [Dataset 9](#), [Dataset 10001](#)
- **Technology Used:** [RDF](#), [SPARQL](#), [SparqlProxy](#), [Google Visualization API](#), [Simile Exhibit](#), [Google Visualization Annotated Timeline](#)
- **Creator(s):** [Dominic DiFranzo](#), [Li Ding](#)

Demo: Comparing Types of Campaign Money by State



- **Description:** Compare the disbursements, receipts, and loans of Democratic and Republican candidates by state. See how these variables are or aren't correlated.
- **Live Demo(s):** [\[3\]](#)
- **Dataset(s):** [Dataset 1930](#), [Dataset 10011](#)
- **Technology Used:** [RDF](#), [SPARQL](#), [SparqlProxy](#), [Google Visualization API](#), [TDB Triple Store](#), [State Variable Comparison JS API](#)
- **Creator(s):** [Sarah Magidson](#)

Demo: Comparing US-USAID and UK-DFID Global Foreign Aid



- **Description:** This application presents a mashup of foreign aid data (represented in US Dollars) from the United States Agency for International Development (USAID) and UK Department for International Development (DFID).
- **Live Demo(s):** [\[4\]](#)
- **Dataset(s):** [Dataset 1554](#), [Dataset 10030](#), [DBpedia](#)
- **Technology Used:** [RDF](#), [SPARQL](#), [SparqlProxy](#), [Google Web Toolkit](#), [Google Visualization API](#), [Google Visualization GeoMap](#), [Google Visualization Pie Chart](#), [How to use New York Times Arc42](#)
- **Creator(s):** [James Michaelis](#)

Demo: Estimate Correlations between Smoking Rate, Cigarette Tax and Beyond

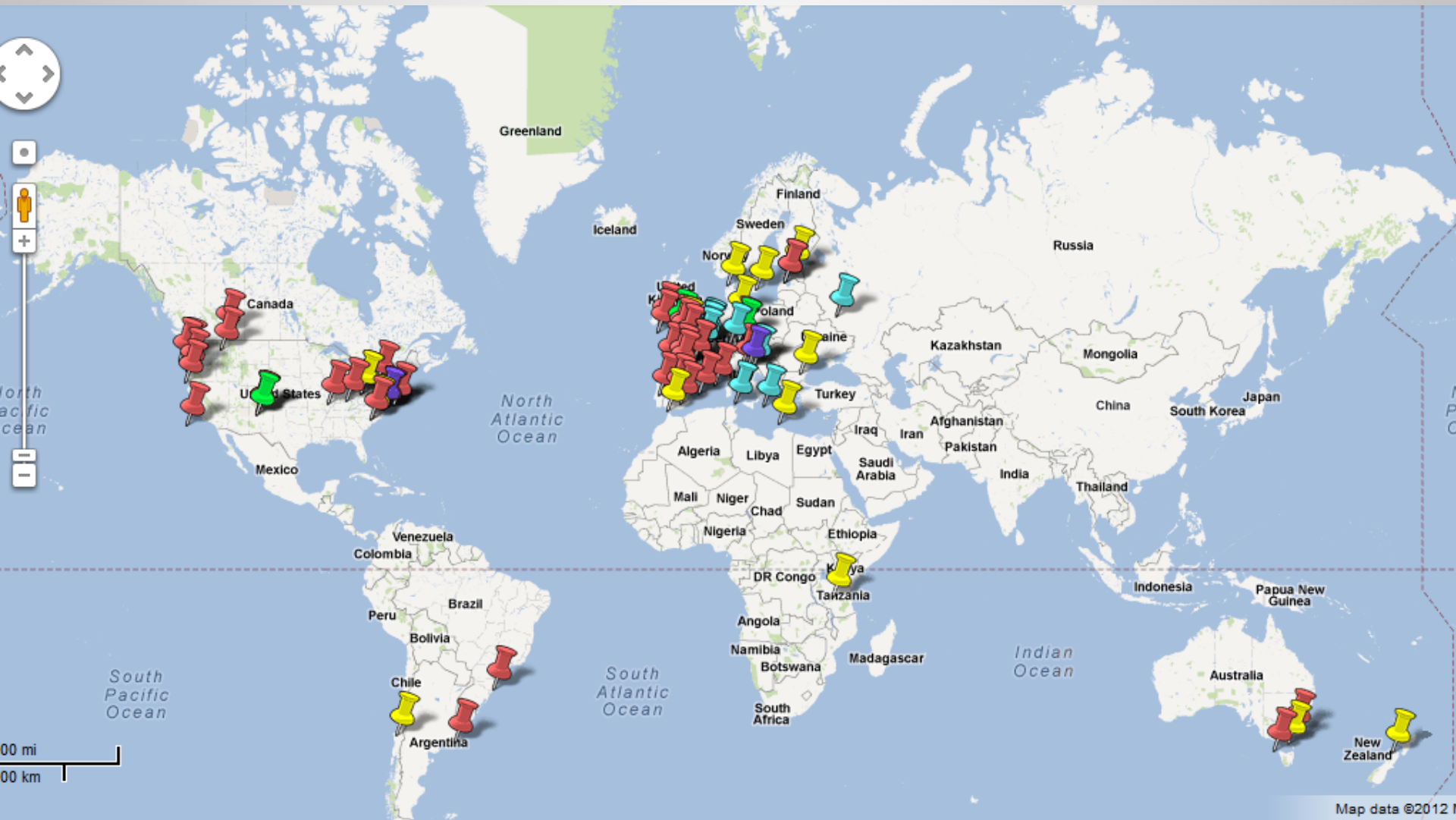


- **Description:** Look at how smoking rates, population, cigarette taxes, and other related variables relate to one another, by state
- **Live Demo(s):** [\[5\]](#)
- **Dataset(s):** [Dataset 353](#), [Dataset 10026](#), [Dataset 10028](#)
- **Technology Used:** [RDF](#), [SPARQL](#), [SparqlProxy](#), [Google Visualization API](#), [TDB Triple Store](#), [State Variable Comparison JS API](#)
- **Creator(s):** [Sarah Magidson](#)

Demo: Faceted browsing on data.gov dataset catalog

Open Data for Open Government

Semantic, Linked, Open



Open Data for Open Government

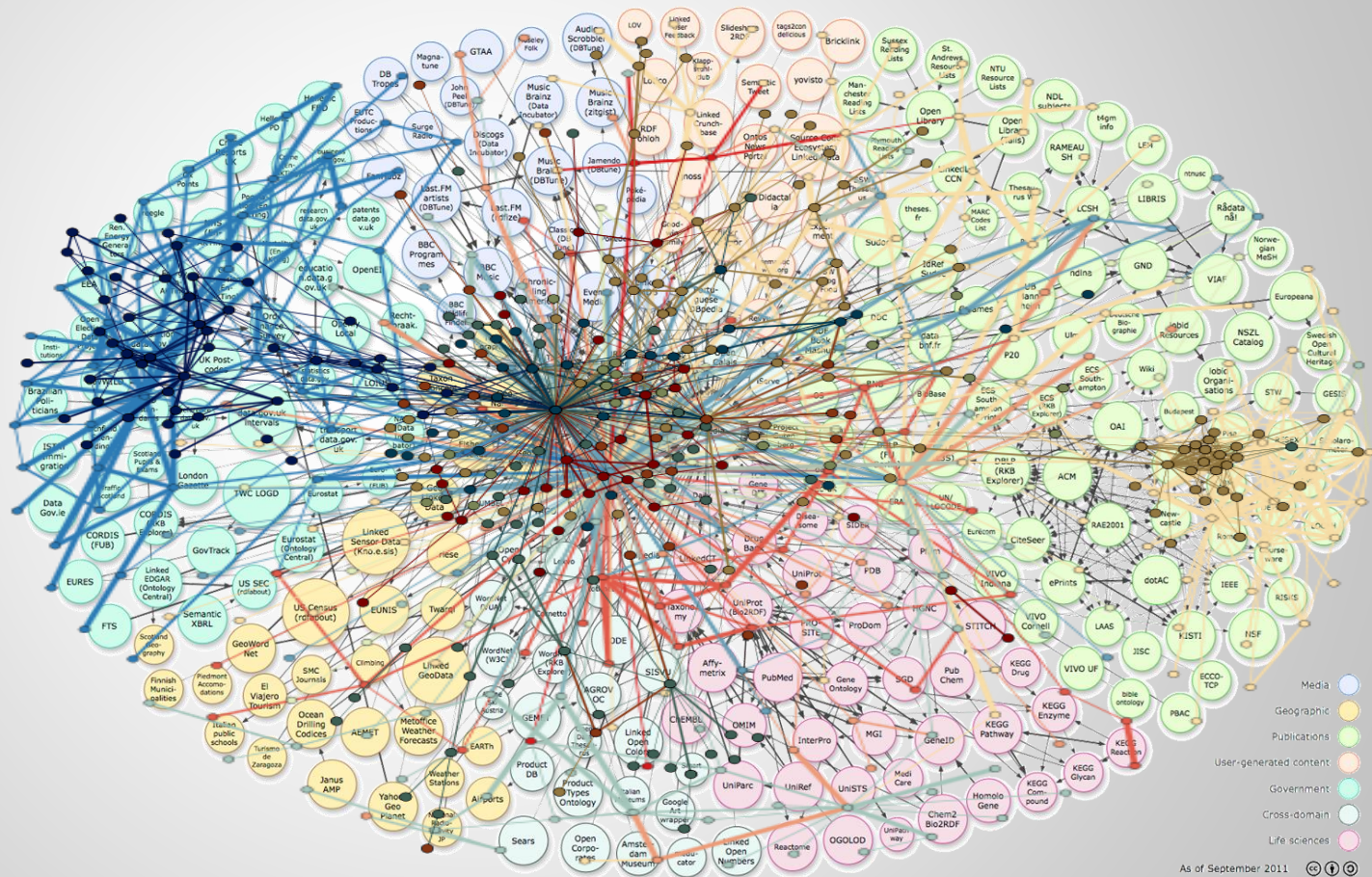
Semantic, Linked, Open



New Research challenges in Knowledge Representation

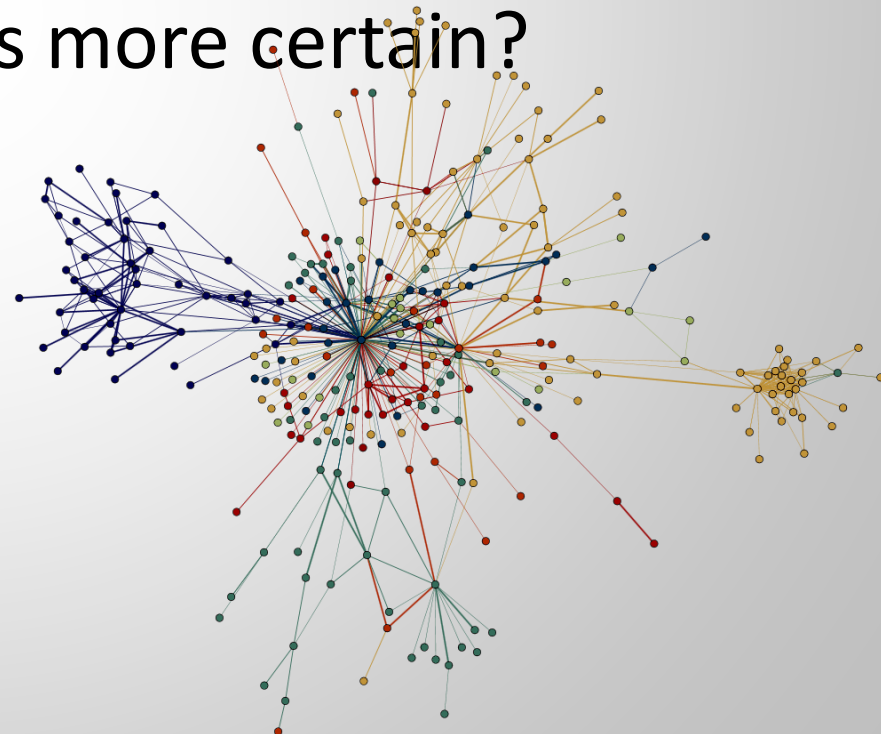
Sentences or network?

$\alpha_1 \rightarrow \beta_1$
 $\alpha_2 \rightarrow \beta_2$
...
 $\beta_n \rightarrow \gamma_n$



What is this picture telling us?

- Does the meaning of a node depend on the cluster it appears in?
- Does path-length correlate with semantic distance?
- Are highly connected nodes more certain?
- Mutual influence of low-level and high-level structure?



How explosive is inference?

Dataset	Input (Triples)	Output (Triples)	Runtime (h:m:s)
Falcon	32.5M	863.7M	0:04:19
Swoogle	78.8M	1.50B	0:07:15
BigWeb	864.8M	30.0B	0:56:57
<i>RDFS</i>			
LDSR	862M	935M	3:31:00
Uniprot	1.5B	2.0B	6:05:49
LUBM	1.1B	496M	0:36:36

OWL-horst

Nobody can
predict these
numbers



Laurens Rietveld
([ISWC 2014](#))

Graph structure as a proxy for semantics

Hotspots in Knowledge Graps

- Observation:
realistic queries only hit a small part of the data (< 2%)
(DBPedia would need 1M queries to hit 2%)

Dataset	Size	#queries	Coverage
DBPedia 3.9	459M	1640	0.003%
Linked Geo Data	289M	81	1.917%
MetaLex	204M	4933	0.016%
Open-BioMed	79M	931	3.100%
Bio2RDF/KEGG	50M	1297	2.013%
SW Dog Food	240K	193	39.438%

Can graph-structure help us here?

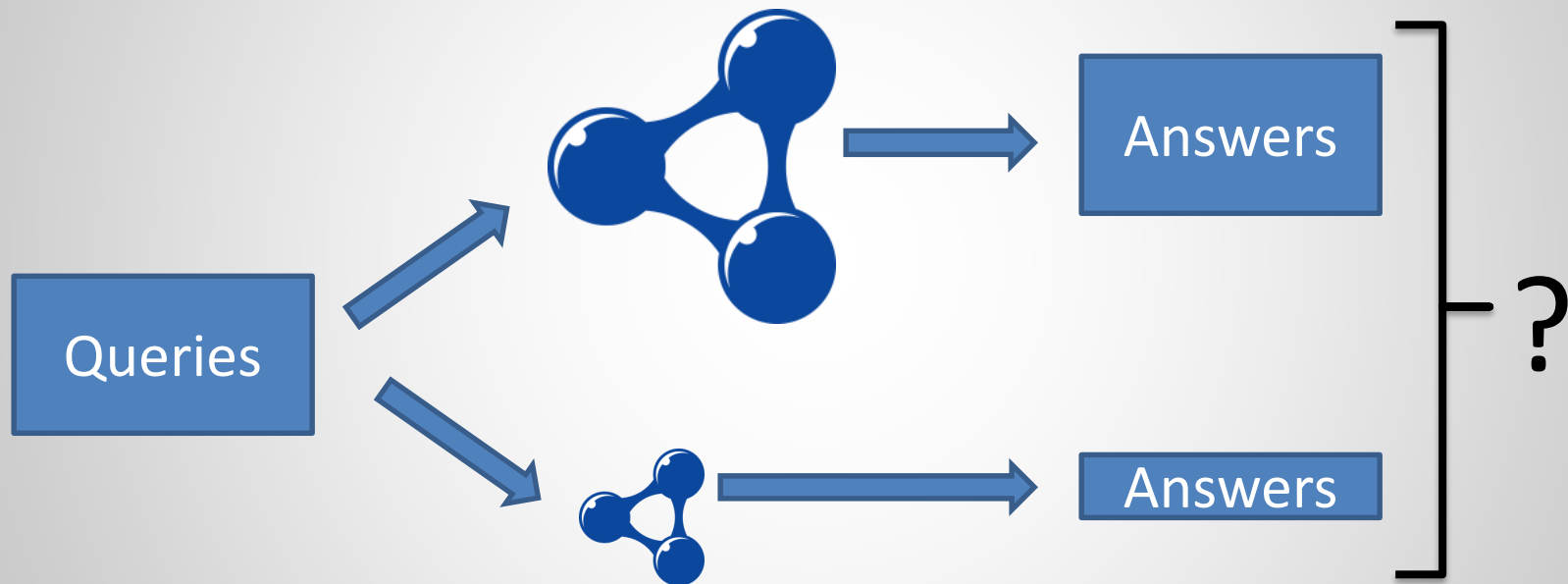
- can we find the popular part of the graph without knowing the queries?

For subgraph $G' \subseteq G$, relevance function $F(G', G)$:
probability that for arbitrary query Q :
 $a \in Q(G) \rightarrow a \in Q(G')$.

- Can we approximate F by simple structural properties?
- = Can we use structure to predict semantic importance?

Experiment

- Can we predict the popular part of the graph *without knowing the queries*?



Use graph-measures for selection:

indegree (easy)

outdegree (easy)

pagerank (doable, iterative & approximate)

betweenness centrality (hard)

Evaluation: example

Dataset			
Subject	Predicate	Object	Weight
:Laurens	:bornIn	:Amsterdam	0.6
:Amsterdam	:capitalOf	:NL	0.1
:Stefan	:bornIn	:Berlin	0.9
:Berlin	:capitalOf	:Germany	0.5
:Rinke	:bornIn	:Heerenveen	0.1

Query

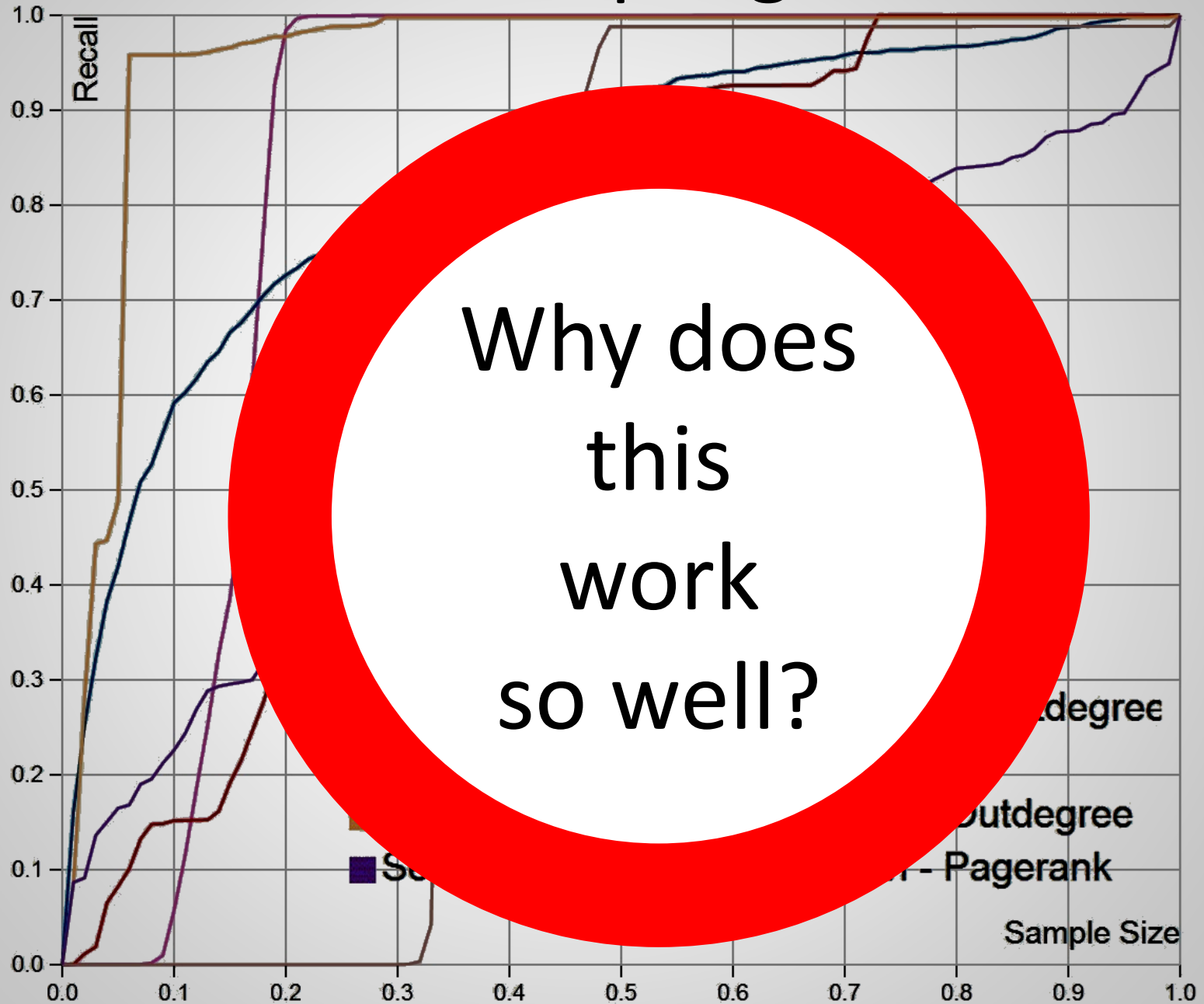
```
SELECT ?person ?country WHERE {  
  ?person :bornIn ?city.  
  ?city :capitalOf ?country  
}
```

Which answers
would we get with
a sample of 60%?



Triples from Query		
Subject	Predicate	Object
:Laurens	:bornIn	:Amsterdam
:Amsterdam	:capitalOf	:NL
:Stefan	:bornIn	:Berlin
:Berlin	:capitalOf	:Germany

Structural sampling: results



Summary

- Two dominant lines of thinking in AI:
 - Statistical
 - Symbolic
- We looked at the symbolic school:
Knowledge Representation
- Many different types of KR
determined by goal, context, trade-offs
- Modern knowledge base can get very large
- They are in widespread practical use
- They generate many open research questions